

MTH 446: Lecture # 1 (catch up)

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Lecture Span

1. Review of complex numbers

Definition 0.1. A set A is a collection of elements combined. They also follow a rule.

Some examples:

1. The empty set $\emptyset$ is the set with no elements
2. If x is an element of A , then we write it as $x \in A$. Else, $x \notin A$.

Following abbreviations, $x \in A \iff x$ is contained in A . Else, x is not contained in A . Some important sets of numbers:

1. $\mathbb{N} = \{1, 2, 3, \dots\}$ (set of Natural numbers)
2. $\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}$ (set of Integers)
3. $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z}\}$ and $\gcd(p, q) = 1$. (set of rational numbers)
4. $\mathbb{R}$, the set of reals

If $A, B \neq \emptyset$, then $y = f(x)$ for $x \in A, y \in B$ is a mapping which assigns each x to a unique y , denoted as $f : A \rightarrow B$, also called a mapping.

If $A, B \neq \emptyset$, then Cartesian product $A \times B$ is $\{(x, y) : x \in A, y \in B\}$

Standard set theory notations for A, B sets:

1. $A \subseteq B \iff x \in A \implies x \in B$
2. $x \in A \cup B \iff (x \in A) \vee (x \in B)$
3. $x \in A \cap B \iff (x \in A) \wedge (x \in B)$
4. $x \in A/B \iff (x \in A) \wedge (x \notin B)$

Rules of addition:

1. Commutative
2. Associative
3. Additive identity
4. Additive inverse

Rules of multiplication

1. Commutative
2. Associative

3. Multiplicative identity
4. Multiplicative inverse
5. Distributive

Let F be a non-empty set on which operations of addition and multiplication are given. Then F is an (algebraic) field if

1. F contains more than one element
2. F with these properties satisfies all addition and multiplication properties.

Thus, we say that $(F, "+", "\cdot")$ is an Algebraic field. As an example, $(\mathbb{R}, "+", "\cdot")$ is an Algebraic field in the field of real numbers.

Hamilton's construction of complex numbers

Recall $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$ and $z = (x, y)$. In complex analysis, define

1. $\mathbb{C} = \mathbb{R}^2$ (complex plane)
2. $z = (x, y)$ (complex number)
3. $x = \text{Re}(z)$ (real part of z)
4. $y = \text{Im}(z)$ (imaginary part of z)

Therefore, in $\mathbb{R}^2$ (and therefore in $\mathbb{C}$), we can define addition as

$$z_1 + z_2 = (x_1 + x_2, y_1 + y_2) \quad (1)$$

Next, introduce multiplication, define

$$z_1 z_2 = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1) \quad (2)$$

Such operations satisfy all rules of addition and multiplication as well as distribution.

Euler's notation

$$z = x + iy.$$